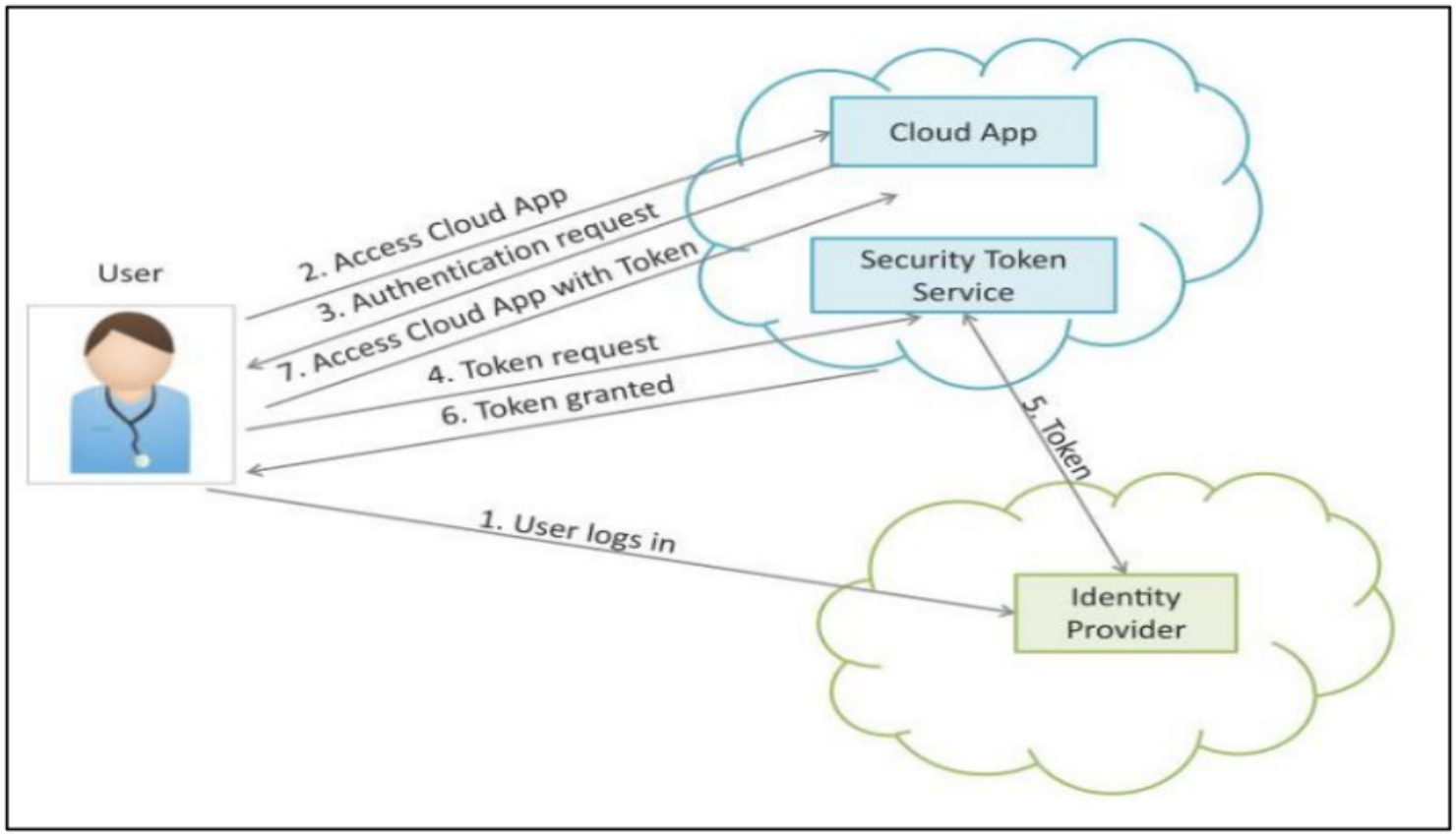
IT Workshop II (Cloud Computing): Cloud Security

October 12, 2026

Authentication

- Security Assertion Markup Language (SAML) is an XML-based open standard data format for exchanging security information (authentication and authorization data) between an identity provider and a service provider.
- SAML-token based SSO authentication



Authentication

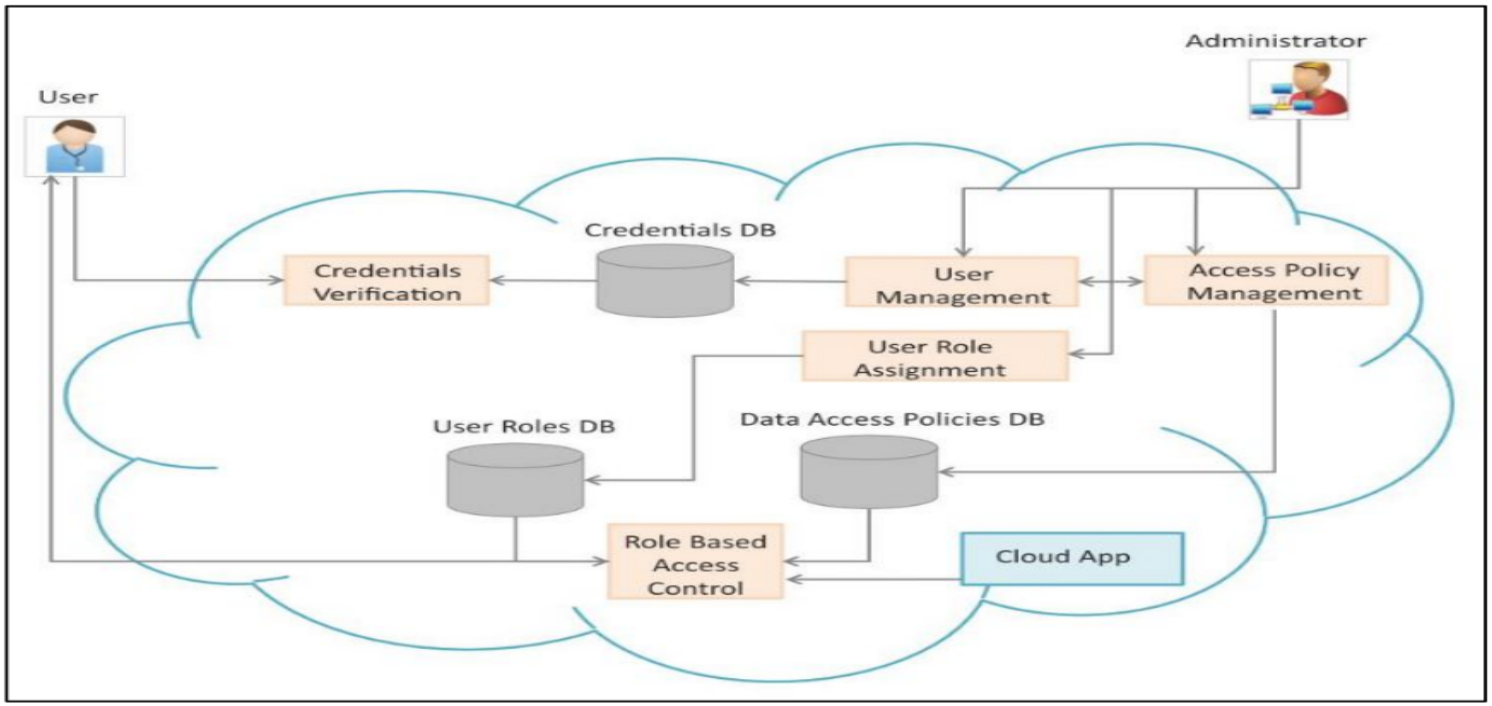
- One time password is another authentication mechanism that uses passwords which are valid for single use only for a single transaction or session.
- Authentication mechanism based on OTP tokens are more secure because they are not vulnerable to replay attacks.
- Text messaging (SMS) is the most common delivery mode for OTP tokens.
- The most common approach for generating OTP tokens is time synchronization.
- Time-based OTP algorithm (TOTP) is a popular time synchronization based algorithm for generating OTPs.

Authorization

- OAuth is an open standard for authorization that allows resource owners to share their private resources stored on one site with another site without handing out the credentials.
- In the OAuth model, an application (which is not the resource owner) requests access to resources controlled by the resource owner (but hosted by the server).
- The resource owner grants permission to access the resources in the form of a token and matching shared-secret.
- Tokens make it unnecessary for the resource owner to share its credentials with the application.
- Tokens can be issued with a restricted scope and limited lifetime, and revoked independently.

Identity and Access Management

- Identity management provides consistent methods for digitally identifying persons and maintaining associated identity attributes for the users across multiple organizations.
- Access management deals with user privileges.
- Identity and access management deal with user identities, their authentication, authorization and access policies.



Securing Data at Rest

- Data at rest is the data that is stored in database in the form of tables/records, files on a file server or raw data on a distributed storage or storage area network (SAN).
- Data at rest is secured by encryption.
- Encryption is the process of converting data from its original form (i.e., plaintext) to a scrambled form (ciphertext) that is unintelligible. Decryption converts data from ciphertext to plaintext.
- Encryption can be of two types:
  - Symmetric Encryption (symmetric-key algorithms)
  - Asymmetric Encryption (public-key algorithms)

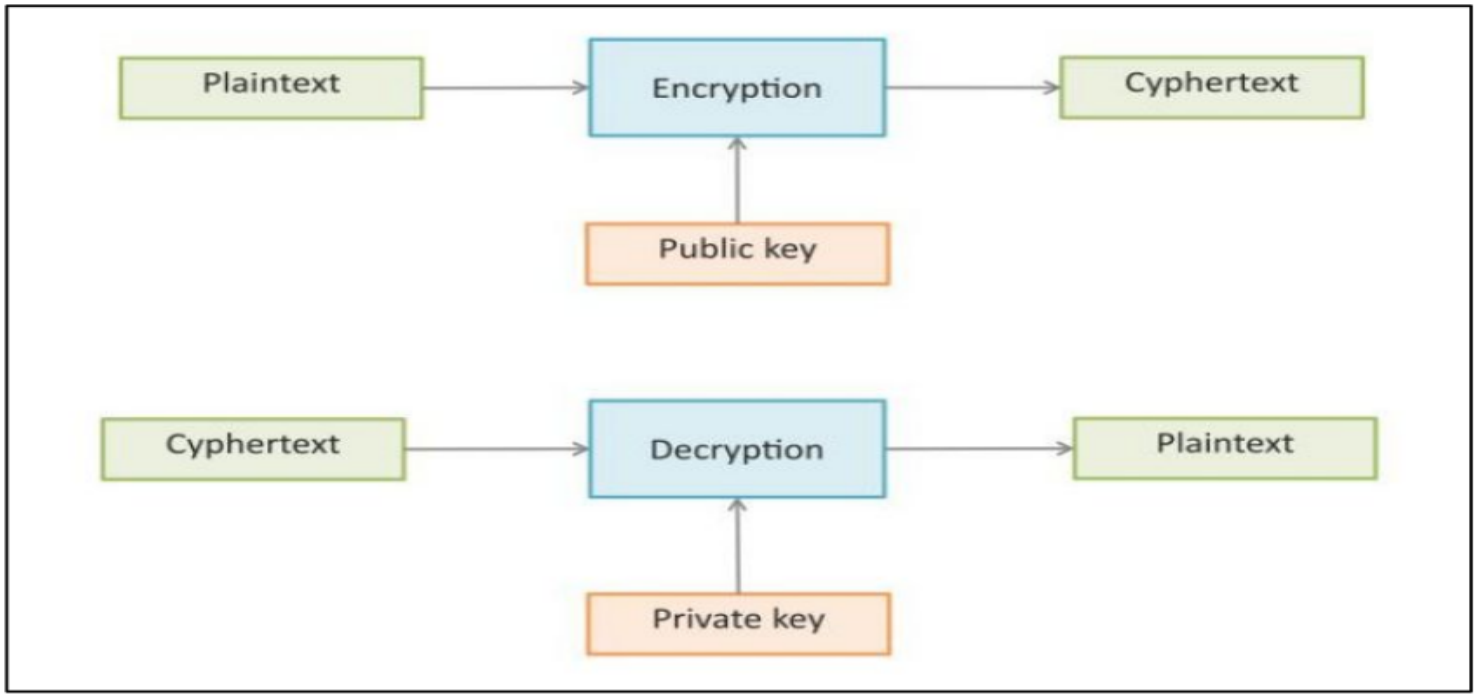
Symmetric Encryption

- Symmetric encryption uses the same secret key for both encryption and decryption.
- The secret key is shared between the sender and the receiver.
- Symmetric encryption is best suited for securing data at rest since the data is accessed by known entities from known locations.
- Popular symmetric encryption algorithms include:
  - Advanced Encryption Standard (AES)
  - Twofish
  - Blowfish
  - Triple Data Encryption Standard (3DES)
  - Serpent
  - RC6
  - MARS

Asymmetric Encryption

- Asymmetric encryption uses two keys, one for encryption (public key) and other for decryption (private key).
- The two keys are linked to each other such that one key encrypts plaintext to ciphertext and other decrypts ciphertext back to plaintext.
- Public key can be shared or published while the private key is known only to the user.
- Asymmetric encryption is best suited for securing data that is exchanged between two parties where symmetric encryption can be unsafe because the secret key has to be exchanged between the parties and anyone who manages to obtain the secret key can decrypt the data.
- In asymmetric encryption a separate key is used for decryption which is kept private.

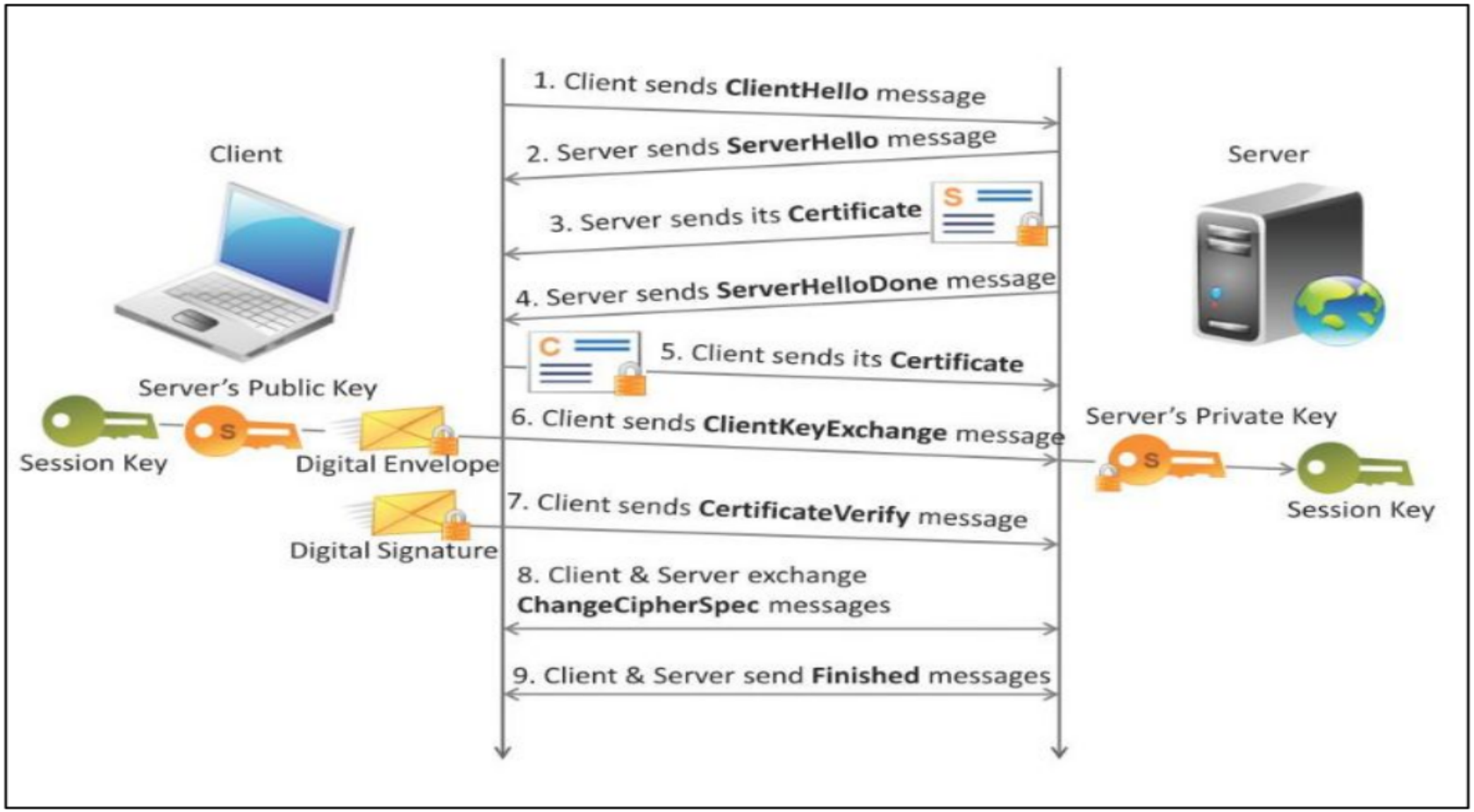
Asymmetric Encryption



Securing Data in Motion

- Securing data in motion, i.e., when the data flows between a client and a server over a potentially insecure network, is important to ensure data confidentiality and integrity.
- Data confidentiality means limiting the access to data so that only authorized recipients can access it.
- Data integrity means that the data remains unchanged when moving from sender to receiver.
- Data integrity ensures that the data is not altered in an unauthorized manner after it is created, transmitted or stored.
- Transport Layer Security (TLS) and Secure Socket Layer (SSL) are the mechanisms used for securing data in motion.
- TLS and SSL are used to encrypt web traffic using Hypertext Transfer Protocol (HTTP).
- TLS and SSL use asymmetric cryptography for authentication of key exchange, symmetric encryption for confidentiality and message authentication codes for message integrity.

Securing Data in Motion



The End